

Windows Event Log Security Analysis Report

Project Information

Project Title: Windows Event Log Security Analysis Lab

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Role: Aspiring SOC Analyst | Cloud Security Professional

Analysis Period: June 2026

Environment: Windows 10

Tools Used:

- Windows Event Viewer
 - Windows Security Logs
 - Log Parser Studio
 - MITRE ATT&CK Framework
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Executive Summary

This project demonstrates the analysis of Windows Security Event Logs to identify, investigate, and document authentication-related security events. The investigation focused on successful and failed logon activities recorded within Windows Security Logs.

Using Event Viewer, authentication events were filtered, analyzed, and correlated to identify patterns that may indicate unauthorized access attempts, password guessing, or brute-force behavior.

The analysis identified multiple failed authentication attempts (Event ID 4625) followed by a successful authentication event (Event ID 4624). While the activity was intentionally generated in a controlled lab environment, the event sequence accurately reflects patterns commonly investigated by Security Operations Center (SOC) analysts.

The project demonstrates practical skills in log analysis, event correlation, threat detection, incident investigation, and MITRE ATT&CK mapping.

Project Objectives

The primary objectives of this project were:

- Access Windows Event Logs using Event Viewer
- Investigate Windows Security Events
- Analyze successful and failed authentication attempts
- Perform event filtering and correlation
- Identify potential indicators of suspicious activity
- Map observed behavior to MITRE ATT&CK techniques
- Document findings using SOC investigation methodology

Lab Environment

Component	Details
Operating System	Windows 10
Log Source	Windows Security Logs
Analysis Tool	Event Viewer
Optional Tool	Log Parser Studio
Investigation Type	Authentication Analysis
Framework	MITRE ATT&CK

Investigation Methodology

Phase 1: Event Log Collection

Windows Event Viewer was used to access:

- System Logs
- Security Logs

Security logs were selected for further investigation due to their relevance to authentication monitoring and incident detection.

Phase 2: Authentication Event Analysis

The Security Log was filtered to investigate authentication-related events.

Event ID 4624

Description:

An account was successfully logged on.

Purpose:

- Verify user authentication activity
 - Track successful access
 - Validate account usage
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Event ID 4625

Description:

An account failed to log on.

Purpose:

- Detect unauthorized access attempts
 - Identify password guessing activity
 - Detect brute-force attacks
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Investigation Findings

Finding 1: Failed Authentication Attempts

Event ID

4625

Severity

Medium

Description

Multiple failed authentication attempts were identified during log review.

Evidence

- Event ID 4625 observed multiple times
- Audit Failure recorded
- Authentication unsuccessful

Failure Details

Field	Value
Event ID	4625
Failure Type	Authentication Failure
Status Code	0xC000006D
Failure Reason	Invalid Credentials
Log Type	Security

Security Impact

Repeated authentication failures may indicate:

- Password guessing
 - Credential attacks
 - Unauthorized access attempts
 - User authentication errors
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Finding 2: Successful Authentication Event

Event ID

4624

Severity

Informational

Description

A successful authentication event was identified following authentication activity.

Evidence

- Event ID 4624 recorded
- Audit Success logged
- User authentication completed successfully

Security Impact

Successful logons should be reviewed when preceded by multiple failed authentication attempts.

Event Correlation Analysis

A timeline review was conducted to determine relationships between authentication events.

Timeline

Time	Event ID	Description
18:12:41	4625	Failed Logon
18:12:44	4625	Failed Logon
18:12:45	4625	Failed Logon
18:13:26	4624	Successful Logon

Correlation Pattern

4625 → Failed Logon

4625 → Failed Logon

4625 → Failed Logon

4624 → Successful Logon

Analyst Assessment

The observed sequence demonstrates multiple authentication failures followed by successful authentication.

In a production environment, this pattern may indicate:

- Password guessing activity
- Brute-force attempts
- User password entry errors
- Credential validation attempts

No evidence of privilege escalation, malware execution, or account compromise was observed during this investigation.

MITRE ATT&CK Mapping

Technique 1

Brute Force

ATT&CK ID: T1110

Description

Adversaries may attempt to gain access by systematically attempting passwords against one or more accounts.

Evidence

Multiple Event ID 4625 entries observed.

Technique 2

Password Guessing

ATT&CK ID: T1110.001

Description

Adversaries attempt to guess passwords for valid user accounts.

Evidence

Repeated failed authentication attempts detected.

Technique 3

Valid Accounts

ATT&CK ID: T1078

Description

Adversaries may use legitimate credentials to gain access to systems and services.

Evidence

Successful authentication event (4624) identified after authentication attempts.

Risk Assessment

Risk Category	Rating
Credential Attacks	Medium
Account Compromise	Low
Privilege Escalation	Low
Malware Activity	None Observed
Overall Risk	Low

Detection Opportunities

Security teams should monitor:

- Excessive Event ID 4625 occurrences
 - Multiple failed logons followed by successful authentication
 - Authentication activity outside business hours
 - Privileged account logons
 - Authentication activity from unusual locations
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Security Recommendations

Authentication Controls

- Enable Multi-Factor Authentication (MFA)
 - Enforce strong password policies
 - Implement password expiration requirements
 - Require complex passwords
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Monitoring Controls

- Monitor Event IDs 4624 and 4625
 - Configure authentication alerts
 - Centralize logs in a SIEM platform
 - Implement security dashboards
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Account Protection

- Enable account lockout policies
 - Review inactive accounts regularly
 - Monitor privileged accounts
 - Audit user access permissions
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SOC Analyst Skills Demonstrated

This project demonstrates practical experience with:

- Windows Event Log Analysis
 - Security Event Monitoring
 - Authentication Investigation
 - Incident Detection
 - Event Correlation
 - Threat Hunting Fundamentals
 - Security Reporting
 - MITRE ATT&CK Mapping
 - Risk Assessment
 - SOC Investigation Methodology
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Conclusion

The Windows Event Log Security Analysis Lab successfully demonstrated the process of identifying, analyzing, and investigating authentication-related security events using Windows Security Logs.

The investigation identified multiple failed authentication attempts followed by successful authentication activity. Event correlation techniques were used to reconstruct the sequence of events and assess potential security implications.

Observed activity was mapped to MITRE ATT&CK techniques including T1110 (Brute Force), T1110.001 (Password Guessing), and T1078 (Valid Accounts), providing threat-informed context to the investigation.

This project reflects the fundamental skills required of a SOC Analyst, including log analysis, event investigation, security monitoring, and incident documentation.